Off-market prediction

90-second speaker script · Member B

188 written words. Rehearse at about 130 words per minute, with brief pauses at slide changes.

Slide 1 — My approach | 0:00–0:38

My part is off-market prediction: estimating a home's sale price without its asking price. I used histogram gradient boosting to learn log sale prices. Learning log prices reduces the influence of very expensive homes. The model captures nonlinear relationships between a home's characteristics and price. It combines size, rooms, parking, property type, synthetic location, listing date, and description cues. It excludes list price and post-sale information. Missing measurements remain unknown, and category encoders learn only from the training fold.

Slide 2 — Measured results | 0:38–1:05

The first graph compares median absolute percentage error. Our model achieved eight point five three percent, against twenty point eight four percent for the nearest-sales reference. Mean absolute percentage error was eleven percent. We tested ten thousand and sixty-nine held-out homes, predicting each month from December twenty twenty-five through May twenty twenty-six using only earlier sales.

Slide 3 — Performance over time and final output | 1:05–1:30

The monthly graph shows error across those six months. I then refitted the model on all one hundred and fifty thousand original sales and generated three thousand nine hundred and seventy-four predictions. Every test ID appears once, with positive, finite prices. These checked predictions provide a reproducible valuation reference. Hidden-test accuracy remains unknown.

Timing note: your 90 seconds leaves 3 minutes 30 seconds for the rest of the five-minute team video.